Kaung Khant Lin

6540131

542

Worksheet 12

Answers

1. (N+1) x (M+1)
2. -
3. -
4. No, it is not worthwhile to DFS further from this state.
5. -
6. -
7. No, it would not be worthwhile. Because the "optimistic estimate" represents the absolute best-case scenario (the absolute maximum money you could possibly make if you broke the rules and perfectly filled every gap).
8. The Bound function calculates the "liquid" estimate (Fractional Knapsack) by greedily grabbing the items starting from index i and filling the remaining capacity C.
9. -
10. -
11. The main idea is to calculate the 'optimistic estimate' (or 'bound') for both the 'Take' and 'Skip' options in the recursion. Then, just check which one has a higher bound and jump into that branch first. It's basically Best-First Search. Because by chasing the branch with the highest upper bound, we're super likely to find a really good, maybe even optimal, item combo early on. That quickly boosts our global best score, let's call it maxV. Since our main way to cut corners (pruning) is optimistic_estimate <= maxV, getting a high maxV fast lets us aggressively ditch tons of weaker, low-potential branches way sooner. This seriously cuts down on the total recursive calls and makes the whole thing run way faster.

Recursion Count Comparison

Test cases	DFS - Not Pruning	Pruning	B & B
0.in	63	41	10
1.in	2097151	612839	72
2.in	65535	44448	105
3.in	33554431	17770698	552
4.in	2047	249	18
5.in	-	-	266
6.in	-	-	506
7.in	-	-	201